MARUTI SUZUKI

# XCELerate 2026

Powering the Future of Mobility!









Platform Partner unstop

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XCELerate 2026

# THESIS.

## PROBLEM STATEMENT

*DESIGN THE IDEAL RIDE-SERVICE VEHICLE FOR INDIA 2035. DETERMINE THE OPTIMAL VEHICLE SIZE, SEATING CONFIGURATION, BODY STYLE, POWERTRAIN STRATEGY AND DIGITAL ECOSYSTEM FOR INTRA-CITY, INTER-CITY, SHARED AND PREMIUM MOBILITY SERVICES. THE SOLUTION SHOULD MAXIMIZE CUSTOMER SATISFACTION, DRIVER EARNINGS, FLEET PROFITABILITY AND SCALABILITY ACROSS INDIA.*



## THE FINDING

Volume is in intra-city. Margin is in airport and commute. They are structurally incompatible — 95 km/day unpredictable hail cannot share a vehicle design with 210 km/day pre-booked depot returns. We did not design for both. We designed for the margin segment.



## THE BUYER

Fleet operators. Every commercial conversation ends at an operations manager's desk and that person buys on cost per km and utilisation. Under ₹67/day flat fee, revenue is fixed — only paid kilometres move the number. That makes uptime and depot control more important than range or features.



## THE DESIGN

Sedan: airport fare premium + luggage requires separate boot. B-segment: sub-4-metre keeps GST at 5%, saves ₹1.5L. BEV on LFP: at 200 km/day return-to-base, depot overnight charging cuts energy cost below every alternative. 45 kWh: sized for the duty cycle with 10% reserve at year 8. Not a better taxi — a fleet asset engineered for this exact operating profile.

MACH1

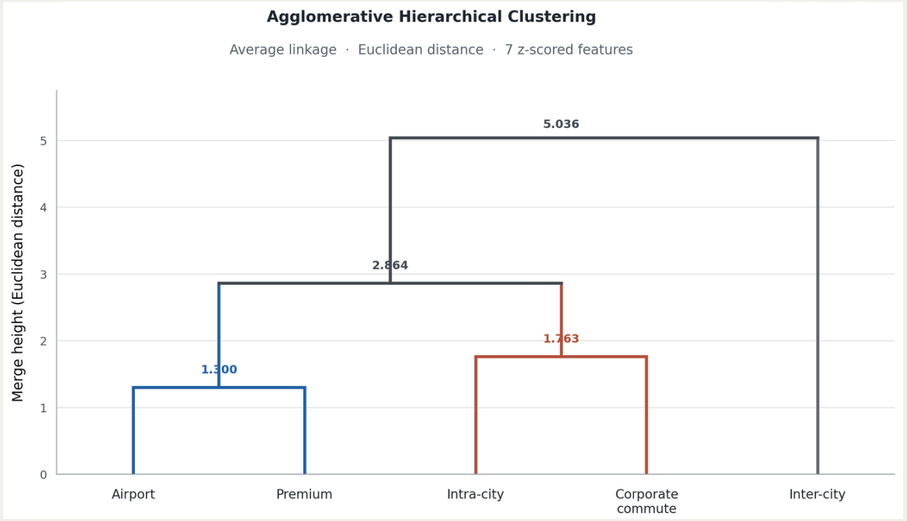
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# MARKET

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## WHAT DOES THE MARKET SAY?

| USE CASE | AVG KM | KM/ DAY | TRIPS/ DAY | ₹/TRIP | GROWTH | EV SHARE | PRICE SENS. |
|---|---|---|---|---|---|---|---|
| Intra-city street hail | 12.7 | 95 | 7.5 | 350 | 18% | 10% | 0.85 |
| Daily commute / subscription | 22.0 | 176 | 8.0 | 550 | 24% | 12% | 0.90 |
| Airport transfer | 35.0 | 210 | 6.0 | 600 | 26% | 18% | 0.35 |
| Inter-city / outstation | 280.0 | 340 | 1.5 | 4,200 | 20% | 5% | 0.40 |
| Premium / shared-premium | 18.0 | 144 | 8.0 | 520 | 28% | 15% | 0.30 |




## WHO IS OUR AUDIENCE?

### FLEET OPERATORS.

*Every commercial conversation about a fleet vehicle ends at an operations manager's desk, and that person buys on cost per kilometre and utilisation.. Structurally, drivers now choose the platform; operators choose the vehicle.*

***BluSmart's exit redistributed supply toward high-utilisation,depot-controlled routes.***

**REPLACEMENT CAPACITY TARGETED AIRPORT AND CORPORATE TRAVEL:**

1. EVERA EXPANDED AT DELHI TERMINALS
2. GMR HYDERABAD LAUNCHED A 200+ VEHICLE PREMIUM AIRPORT FLEET
3. BHARAT TAXI LAUNCHED AT NAVI MUMBAI.

| RANK | CRITERION | EVIDENCE |
|---|---|---|
| 1 | Cost per kilometre | every fleet TCO source converges here |
| 2 | Paid km/day (utilisation) | now the binding constraint under flat-fee platform economics |
| 3 | Uptime | charging downtime is the failure mode operators actually fear |
| 4 | Depot control | without controlled overnight charging, EV TCO does not hold |
| 5 | Residual value | BluSmart assumed 10%, realised 26% — our exact target segment |
| 6 | Financeability | CAPEX structure alone is worth ~8 points of EV advantage |
| 7 | Range | near the bottom — 315 km against a 200 km/day duty is ample |

## WHERE DO WE PITCH?

*India's ride-service **volume** is in intra-city.*
*India's ride-service **margin** is in airport and commute.*
*This matters because they require fundamentally **different** vehicles.*

| USE CASE | KM/DAY | GROWTH | EV SHARE | DRIVER ₹/MO | MACH1 FIT |
|---|---|---|---|---|---|
| intra-city | 95 | 18% | 10% | 68,250 | weak |
| premium | 144 | 28% | 15% | 108,160 | moderate |
| commute | 176 | 24% | 12% | 114,400 | STRONG |
| airport | 210 | 26% | 18% | 93,600 | STRONG |
| intercity | 340 | 20% | 5% | 163,800 | weak |

**THE PRIMARY POSITION**
**CLUSTER A — AIRPORT + CORPORATE-COMMUTE**

- *TRIPS ARE PRE-BOOKED → THE VEHICLE'S DAY CAN BE PLANNED*
- *PICKUP IS A FIXED POINT → DEPOT CHARGING IS SCHEDULABLE OVERNIGHT*
- *THE DUTY IS PREDICTABLE → THE 45 KWH PACK AND THE 200 KM/DAY SIZING ARE SAFE*
- *EV SHARE IS ALREADY 18% → THE OPERATOR IS ALREADY ELECTRIFYING THIS LANE*

**AIRPORT TRANSFERS ARE ALSO THE FASTEST-GROWING TRIP PURPOSE AT 19.14% CAGR, AND UBER BLACK'S TOP TWO USE CASES ARE AIRPORT AND CORPORATE TRAVEL.**

**WHERE WE DO NOT FIT**

**INTRA-CITY**

- 61% OF THE MARKET, BUT WEAKEST FIT FOR E1.
- 95 KM/DAY; UNPREDICTABLE HAIL COMPLICATES DEPOT SCHEDULING.
- HIGHEST PRICE SENSITIVITY.

**INTER-CITY**

- 560 KM DRIVEN PER ROUND TRIP FOR 280 KM OF REVENUE.
- LOWEST EV SHARE: 5%.
- EMPTY RETURN TRIPS REDUCE EFFECTIVE UTILISATION AND WEAKEN ECONOMICS.

# VEHICLE CONCEPT PROPOSAL

## Powertrain & Energy

| FLEETLINE E1 FEATURE | WHY A FLEET OPERATOR NEEDS IT |
|---|---|
| SPEC 45 kWh LFP pack, semi-structural CTP | 200 km/day duty with 10% reserve held at year 8 |
| SPEC PMSM 80 bhp, single-speed reduction drive | No shift loss across 62,400 km/yr, and no gearset to maintain |
| SPEC Liquid-plate cooling, 48 °C cell design point | Holds degradation inside the validated band — passive cooling does not |
| ADD ⚑ Per-cell pack SoH telematics, over-the-air | DECISIVE — THE ONE THAT EARNS THE WARRANTY. Published LFP studies disagree 3.6× on time-to-80%, so a capacity-retention guarantee is only credible if the pack reports its own degradation |

## Safety & Liability

| FLEETLINE E1 FEATURE | WHY A FLEET OPERATOR NEEDS IT |
|---|---|
| SPEC L2 ADAS, six airbags, ABS with EBD + CSC | Corner Stability Control matters more on a tall, pack-loaded sedan |
| ADD 360° HD camera with 2D/3D views | Blind-spot elimination at low speed, and dense-city parking without a spotter |
| ADD Dashcam recorder (DVR) | 10+ paid hours a day — onboard evidence for disputes, when the driver's account is not enough |

## Uptime & Reliability

| FLEETLINE E1 FEATURE | WHY A FLEET OPERATOR NEEDS IT |
|---|---|
| ADD Intelligent tyre pressure monitoring | Turns a roadside blowout into a scheduled wheel change. Tyre failure is a leading cause of lost paid hours |

## Driver Working Conditions

| FLEETLINE E1 FEATURE | WHY A FLEET OPERATOR NEEDS IT |
|---|---|
| ADD 65 W USB-C front + rear wireless phone deck | Two riders, two phones, one shift, no cable hunt between fares |
| ADD Ventilated front seats | A 12-hour shift is the working condition, not a luxury — heat and fatigue show up as unpaid kilometres |

## LEGEND

| | |
|---|---|
| SPEC | Already specified — engineering locked |
| ADD | Being added — requested by this proposal |
| ⚑ / DECISIVE | The line that converts a spec sheet into a commercial argument |

| Parameter | FleetLine E1 |
|---|---|
| Body Style | Compact Sedan |
| Segment | B-segment |
| Length | 3,995 mm |
| Width | 1,684 mm |
| Height | 1,532 mm |
| Wheelbase | 2,450 mm |
| Seating | 5-seat, 2-row |
| Boot Space | 400 L ↑ Tour S |
| Powertrain | 45 kWh BEV |
| Fleet Pricing | ~₹14 L (EV) |

***THE DZIRE TOUR S IS INDIA'S GOLD-STANDARD FLEET SEDAN*** *5,000 UNITS/MONTH, 40% OF ALL DZIRE SALES, OPERATORS RETURN TO IT EVERY TIME. FLEETLINE E1 DOES NOT REINVENT THIS FORMULA. IT ELECTRIFIES IT.*

### 🏆 *PROVEN IN INDIAN FLEET — MARKET EVIDENCE*

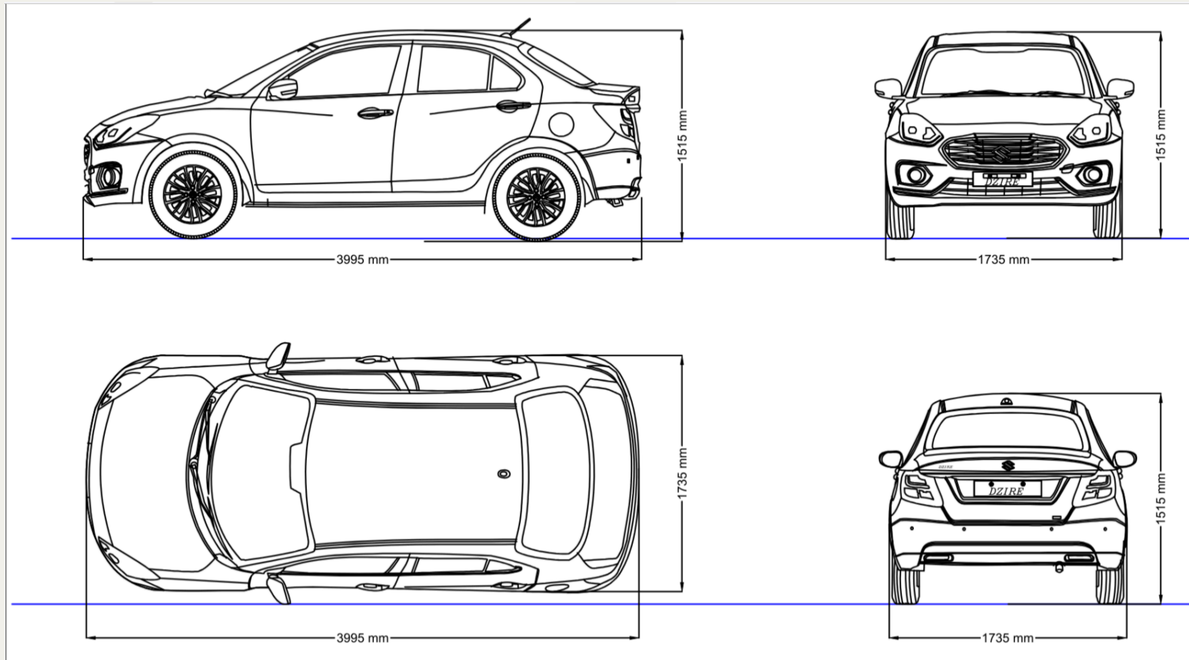
- *RETURNED TO TOP OF SALES CHARTS THE MOMENT MARUTI REOPENED FLEET REGISTRATIONS — OPERATORS RESPONDED IMMEDIATELY*
- *40% OF ALL DZIRE UNITS GO TO FLEET BUYERS — 5,000 UNITS/MONTH AT STEADY STATE*

### 📐 *WHY THESE EXACT DIMENSIONS WORK*

- *3,995 MM SUB-4-METER KEEPS GST AT 5% (VS 18% FOR ≥4 M) — STRUCTURAL COST ADVANTAGE OF ~₹1.5 L*
- *2,450 MM WHEELBASE SEATS ADULTS WITHOUT DRAMA AND FITS EVERY AIRPORT BAY AND CITY LANE*

### 🔧 *WHY OPERATORS CHOOSE IT OVER RIVALS*

- *MARUTI'S 3,614-OUTLET SERVICE NETWORK MEANS A CAB IS NEVER MORE THAN A FEW KM FROM A SERVICE POINT*
- *SPARE PARTS STOCKED EVERYWHERE — NO WAITING, NO IMPORTED COMPONENTS, NO DOWNTIME*



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# TECHNICAL SPECIFICATIONS

| | |
|---|---|
| **POWERTRAIN** | 80 bhp continuous / 100 bhp peak - PMSM - Water-cooled<br>*Peak + short duration, continuous rating derates at 45°C ambient* |
| **DRIVE** | Single-speed reduction - BEV |
| **MOTOR EFFICIENCY** | 93–97% peak - –2% drop at 45°C |
| **BATTERY / RANGE** | LFP - 45 kWh - Structural pack<br>10–90% - 36 kWh usable<br>*3% top and bottom reserved for cycle-life protection* |
| **RANGE** | 263 km claimed - 200 km @ 75% DoD (feet (operating range) |
| **DAILY ENERGY** | –25.7 kWh @ 200 km/day - –20–35% with AC load at 45°C |
| **PACK THERMAL MGMT** | Active liquid cooling - pack maintained ≤45°C |
| **CHARGE RATE** | 1C max - 45 kW DC fast charge - CC–CV protocol |
| **DEPOT CHARGING** | 11 kW AC - 3-phase - –23 hr overnight window |
| **RATED LIFE** | 200,000+ km - –3.2 yr at 62,400 km/yr duty cycle |
| **BRAKING / CHASSIS** | Front: Disc - Rear : Drum<br>ABS - ESC |
| **WHEELS / TIRES** | 14' alloy - 175/56 R14 tubeless radial |
| **GROUND CLEARANCE** | 170 mm unladen |
| **SAFETY / COMPLIANCE** | **Airbags - ABS - Level 2 ADAS**<br>**AIS-156 - AIS-038 Rev.2 - ECE R100** |
| **BMS** | Cell-level voltage + temperature monitoring |
| **INTERIOR / COMFORT** | 5-seat - 2row - rear-biased - flat floor (no tunnel)<br>Rear AC vents - durable leatherette - ventilated front seats<br>EPS - Full LED - DRL bar - Fog lamps |
| **TOP VARIANT** | Panoramic sunroof |
| **LUGGAGE** | 400 L |
| **BATTERY FIRST-LIFE** | – 15 yr (cycle-life basis) |
| **CORROSION / NVH** | Validated for coastal and monsoon deployment |
| **PRIMARY WEAR ITEM** | Bearings |
| **SEGMENT** | Compact Sedan - B-segment |

| 3,995 | 1,684 | 1,532 | 2,450 |
|---|---|---|---|
| LENGTH MM | WIDTH MM | HEIGHT MM | WHEELBASE MM |

# POWERTRAIN STRATEGY

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## 01 — COMPARISON AND JUSTIFICATION

| | BEV | HEV | PHEV | CNG | BIO CNG |
|---|---|---|---|---|---|
| COST/KM (OPERATOR) | ₹6.42 | ₹9–11 | ₹8–10 | ₹7.42–8.34 | ≈ CNG + supply risk |
| REFUELLING READINESS | Depot AC overnight; zero revenue-hour downtime | Full — ICE base | Full — ICE base | Best-in-class | Best-in-class |
| RANGE / UPTIME | 315 km real (verified) | Unaffected | Unaffected | ~500–600 km real | ~500–600 km real |
| SUSTAINABILITY | Best available on a coal-heavy grid | Marginal | Marginal | Methane slip; worse CO2 | Genuinely low-carbon but supply immature |
| CHOSEN BY REGULATORS | Yes — Delhi gazette 100% by 2030 | No | No | No | No |

*India has roughly 35,000 public charging ports for 5.7 million registered EVs — 160 EVs per port (NITI Aayog index, via Cadmus/Uber). Public charging is therefore not an operating plan for a fleet. It is a fallback.*

## WHY NOT THE ALTERNATIVES?

*BEV OFFERS THE LOWEST OPERATING COST FOR OUR 200 KM/DAY DEPOT-BASED DUTY CYCLE, WITH CHEAPER ENERGY, LOWER MAINTENANCE, AND OVERNIGHT CHARGING THAT AVOIDS REVENUE LOSS. LFP KEEPS THE BATTERY COST-EFFECTIVE AND DURABLE.*

### CNG — THE REAL COMPETITOR, AND IT LOSES ON COST

*CNG wins refuelling readiness and has a far lower purchase price. It loses because at 200 km/day the running-cost saving is not enough to cover the ₹5.49 lakh capital premium within five years, after which EV wins for the remaining life.*

### BIO-CNG — THE OPTION THAT IS NOT READY

***Bio-CNG has the best carbon story of any fuel available, and it is not deliverable at fleet scale in 2035:***

- *207 CBG plants commissioned as of April 2026, against a government target of 5,000 by 2030*
- *Only 17 of those 207 are pipeline-connected — the rest move by high-pressure cascade*
- *Commissioned capacity is ~2 MMTPA against a 15 MMTPA target*
- *Operating plants run at 20–60% of rated capacity on feedstock inconsistency*
- *Mandatory blending is only 1% in FY2025-26, rising to 5% by FY2028-29*

### HEV AND PHEV — STRUCTURALLY UNCOMPETITIVE HERE

*CEEW is unambiguous:*
***"Petrol-electric hybrid vehicles... are significantly more expensive, mainly due to the high GST and compensation cess (at 43 per cent), making them uncompetitive in this segment against EVs."***
*The tax treatment alone is disqualifying. Beyond that, both inherit the ICE's full mechanical service burden — the exact cost line where EV gains its advantage — while adding a battery and inverter to a fleet that will keep both.*

## BEV. LFP. SINGLE-SPEED DRIVE.

High-mileage taxis favour EVs on total cost of ownership, with CNG as the alternative where infrastructure exists. — CEEW

**200 km/day makes the economics work.**

Fleet parity with diesel typically emerges at 120–200 km/day, with 18–24-month payback. Our depot-return duty cycle sits at the favourable end of this range.

**Depot charging eliminates downtime.**

A 36 kWh recharge (10–90%) takes 4.9 hours at 7.4 kW or 3.3 hours at 11 kW—completed overnight, with zero revenue hours lost.

**Cheap, scheduled charging protects margins.**

Depot electricity at ₹5.50–9/kWh beats public fast charging at ₹18–24/kWh. Smart off-peak scheduling can cut charging costs by 20–35%, with a feasible 1:3 charger-to-vehicle ratio.

## 02 — CHARGING & REFUELING READINESS

### WHY BEV WOULD WORK?

| | TIME FOR 36 KWH (10→90%) | FITS OVERNIGHT WINDOW? |
|---|---|---|
| 7.4 KW AC | 4.9 h | Yes |
| 11 KW AC | 3.3 h | Yes |
| 22 KW AC | 1.6 h | Yes |

| CHARGING ROUTE | COST |
|---|---|
| PUBLIC DC FAST CHARGE | ₹18–24 /kWh |
| DEPOT, OFF-PEAK COMMERCIAL TARIFF | ₹5.50–9 /kWh |
| DEPOT + ROOFTOP SOLAR | ₹3–5 /kWh |

### PACK SIZING HEADROOM

MACH1(EV)

| | |
|---|---|
| 200 KM/DAY | 34 kWh → 45 kWh |
| 263 KM/DAY | 45 kWh |
| 300 KM/DAY | 51 kWh → 45 kWh |

## WHAT ACTUALLY HAPPENS WITHOUT DEPOT CHARGING

*"Around two-and-a-half months after... Green SM launched its electric taxi service... drivers said they are spending hours waiting for chargers or going offline to conserve battery... I sometimes come to charge my vehicle on my week-offs." — The Indian Express, 1 Sep 2026, Noida*

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**A SECOND CHEMISTRY.**
**A SOVEREIGN SUPPLY CHAIN.**
**A CALCULATED BET ON 2035.**

# THE SOVEREIGN'S BET

***THE INDUSTRY HAS A DEFAULT. WE HAVE A DIFFERENT THESIS.***
*SODIUM-ION NEED NOT DEFEAT LITHIUM-ION TODAY. IT NEEDS TO BECOME A CREDIBLE ALTERNATIVE BY 2035. MACH1 IS DESIGNED TO GIVE IT THAT CHANCE—WITH LFP AS THE FALLBACK AND THE FINAL CHEMISTRY DECIDED BY COST, SUPPLY SECURITY AND DEMONSTRATED PERFORMANCE.*

## 1. BATTERY INDEPENDENCE IS STRATEGIC, NOT JUST ECONOMIC.

India's EV supply chain remains heavily dependent on imported lithium-ion cells and materials. **Sodium-ion offers a second chemistry**—reducing lithium dependence, enabling aluminium current collectors, and creating opportunities for domestic supply-chain control. The goal is not to replace LFP, but to reduce reliance on a single chemistry.

## 2. 2035 GIVES THE TECHNOLOGY TIME TO MATURE.

Sodium-ion manufacturing remains small today, but major manufacturers are scaling development, with improved low-temperature performance and growing industrial interest. **MACH1 does not bet on sodium-ion winning; it preserves the option.**

## 3. A STAGED DECISION KEEPS THE RISK CONTAINED.

- **By 2032–33**: Adopt sodium-ion only if it meets cost, performance, and supply-chain requirements.
- **If it falls short:** Launch with LFP, the established fallback.
- **Design principle:** One vehicle architecture, two battery pathways, no dependence on an unproven outcome.

## *PUBLIC FUNDING IS MOVING SODIUM-ION RESEARCH TOWARDS COMMERCIALISATION*

**May 2025 — JNCASR develops a fast-charging sodium-ion cell**

The Department of Science and Technology (DST) publicised research by the Jawaharlal Nehru Centre for Advanced Scientific Research (JNCASR), reporting a sodium-ion battery that charged to 80% in six minutes and achieved more than 3,000 charge cycles in the reported testing.

**December 2025 — ARCI moves into industrial cell validation**

The International Advanced Research Centre for Powder Metallurgy and New Materials (ARCI), an autonomous R&D institution under DST, signed an agreement with Voltasun Technologies on 15 December 2025 to supply 80 sodium-ion pouch cells rated at 5 Ah each for industrial-performance evaluation.

**December 2025 — Government-backed Indo-Israel industrial R&D**

The Technology Development Board (TDB), under DST, entered into an agreement with NOPO Nanotechnologies India under the India–Israel Industrial R&D and Technological Innovation Fund. The project investigates integrating single-walled carbon nanotubes into sodium-ion batteries and developing a proof-of-concept prototype.

**April 2026 — Direct support for commercial-scale sodium-ion materials**

TDB announced financial assistance to Indigenous Energy Storage Technologies Pvt. Ltd. for commercialisation of hard carbon derived from agricultural and other biomass waste, intended for use as a sodium-ion battery anode material.
This is strategically important because hard carbon is a key sodium-ion input, and the IEA identifies its supply chain as underdeveloped and heavily concentrated in China

**9th September 2026 — NTPC issues a sodium-ion grid-storage pilot EOI**

This is one of the clearest current policy signals. NTPC's official tender portal records an Expression of Interest titled "EOI for Setting up A Pilot Project for Development and Installation of a Grid-Connected BESS using Sodium-Ion Battery Technology."

## BROADER NATIONAL POLICY IS CREATING THE CONDITIONS FOR BATTERY-CHEMISTRY DIVERSIFICATION

*TOGETHER, THESE INITIATIVES DEMONSTRATE A NATIONAL EFFORT TO REDUCE DEPENDENCE ON IMPORTED BATTERY INPUTS AND BUILD DOMESTIC MANUFACTURING CAPABILITY. SODIUM-ION IS RELEVANT TO THAT OBJECTIVE BECAUSE IT OFFERS AN ALTERNATIVE PATHWAY WITH A DIFFERENT RAW-MATERIAL PROFILE.*

NATIONAL PROGRAMME ON ADVANCED CHEMISTRY CELL BATTERY STORAGE.

The Ministry of Heavy Industries' PLI scheme has an approved outlay of ₹18,100 crore to build gigascale advanced-chemistry-cell manufacturing in India, with a stated emphasis on domestic value addition. The scheme provides an industrial framework for building battery manufacturing capabilities

NATIONAL CRITICAL MINERAL MISSION

Approved in January 2025, the mission provides for ₹16,300 crore in public expenditure over seven years, with an expected ₹18,000 crore of investment from public-sector undertakings and other stakeholders. Its purpose is to strengthen domestic critical-mineral value chains across exploration, mining, processing and recovery.

ICEA ENERGY STORAGE REQUIREMENTS

India's Central Electricity Authority has estimated an energy-storage requirement of approximately 411 GWh by 2031–32, driven by the need to integrate renewable electricity reliably. The figure was reported in a December 2025 parliamentary response published by the Ministry of Power.

*INDIA IS PURSUING BATTERY SELF-RELIANCE AND CRITICAL-MINERAL DIVERSIFICATION; SODIUM-ION IS EMERGING AS ONE TECHNOLOGY CAPABLE OF CONTRIBUTING TO THOSE GOALS.*

# WHY 2035 DOES NOT SOLVE IT

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*This is not a naive claim, and it deserves a fair hearing. Sodium-ion's cost trajectory is real, and layered-oxide sodium cells genuinely do reach LFP price parity on a five-year view.*
***IT FAILS ON TIMING AND CERTIFICATION, NOT ON DIRECTION.***

## WALL 1 - AVAILABILITY AT SCALE

Current automotive deployments are 23–25 kWh microcars, not sedan-class vehicles. CATL's Naxtra has GWh-scale capacity and passenger-car certification, but uses Prussian White—not the layered-oxide chemistry specified for MACH1. These chemistries are not interchangeable drop-in solutions.
**Decision: Layered-oxide sodium-ion remains a future option, not a launch-ready cell. LFP remains the fallback unless the specified chemistry reaches the required scale, performance, and certification by 2032–33.**

## WALL 2 — IT CANNOT BE LEGALLY HOMOLOGATED IN INDIA.

**AIS-038 REV.2 AMENDMENT 3, ANNEX IX-K(2), VERBATIM:**
"CELLS USED TO MAKE REESS, SHALL BE CERTIFIED AS PER IS 16893-PART 2 AND PART 3 BY NABL ACCREDITED LAB OR BY TEST AGENCY NOTIFIED UNDER CMV RULE 126."
IS 16893-2 AND -3 ARE TITLED "SECONDARY LITHIUM-ION CELLS...". NO AMENDMENT THROUGH AMD 4 HAS ADDED AN ALTERNATIVE-CHEMISTRY CLAUSE OR SCOPE EXEMPTION. NO SODIUM-ION CELL HAS EVER BEEN CERTIFIED TO THEM.

## *DECISION*

***LFP IS THE LAUNCH CHEMISTRY. SODIUM-ION IS THE 2036–38 SECOND PHASE,*** *RETAINED THROUGH A DUAL-CHEMISTRY ARCHITECTURE. THE ISSUE IS NOT SODIUM-ION'S POTENTIAL, BUT ITS READINESS: IT CANNOT MEET THE PROPOSED CERTIFICATION TIMELINE.* ***IF IT CANNOT BE CERTIFIED, IT CANNOT BE DELIVERED.***

## *ARCHITECTURE*

*WHY DETACHABLE, NOT STRUCTURAL*

- *CHEMISTRY SWAP BECOMES A* ***SOFTWARE CHANGE,*** *NOT A NEW CAR*
- ***AIS-038 CLAUSE 2.48:*** *A CHEMISTRY CHANGE NEEDS FULL REESS RETEST EITHER WAY*
- *BUT* ***DETACHABLE MAKES THE CELL SWAPPABLE.*** *STRUCTURAL CTC LOCKS THE CHEMISTRY FOREVER*
- *COST OF THAT OPTIONALITY: ₹2–3 CR / 4–6 MONTHS VS ₹15–25 CR / 18–24 MONTHS*

| DIMENSION | DETACHED SEMI-STRUCTURAL CTP | FULL STRUCTURAL CTC |
|---|---|---|
| Chemistry change | Cell swap + firmware | Effectively impossible |
| AIS-038 Clause 2.48 retest | 4–6 months, 2–3 Cr | 18–24 months, 15–25 Cr |
| Cell standardisation | 256×106×32 prismatic, 102–108s | Chemistry-locked |
| Repurposing / second life | Straightforward | Very difficult |
| Body programme impact | None | Full body-in-white retool |

| Pack sizing | SoH at year 8 | Design cell temp |
|---|---|---|
| 44.2 kWh @ 48°C cell | 77.3% — commercially viable past loan payoff | 48°C max under liquid cooling |

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# TOTAL COST OF OWNERSHIP ANALYSIS

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## 0. METHODOLOGY
## STEP-BY-STEP COST ANALYSIS

1. **Define the operating case:** Compare both vehicles at 200 km/day over five- and eight-year ownership periods.
2. **Collect input data:** Gather vehicle prices, financing terms, energy efficiency, tariffs, maintenance, insurance, taxes and resale values across three cities.
3. **Calculate acquisition cost:** Use the on-road price for cash purchases or total EMI payments for financed purchases, never both.
4. **Calculate energy cost:** Estimate lifetime electricity and CNG expenditure, accounting for charging losses and applicable tariffs.
5. **Calculate maintenance and insurance:** Include servicing, repairs, tyres, insurance and other relevant operating expenses.
6. **Assess uptime and lost earnings:** Evaluate charging and repair downtime and its potential impact on operating revenue.
7. **Account for residual value:** Estimate and deduct the vehicle's resale value at the end of ownership.
8. **Calculate TCO and cost/km:** Combine ownership and operating costs, subtract residual value, and calculate total cost and cost per kilometre.
9. **Compare operator and driver economics:** Evaluate ownership costs, operating expenses and driver take-home earnings separately.
10. **Validate and test sensitivity:** Verify calculations against external sources and test variations in energy prices, efficiency, daily distance, financing, downtime, resale value and battery replacement.

## 2.ECONOMIC BOUNDARY

| INCLUDED | DETAIL |
|---|---|
| ACQUISITION | Ex-showroom, GST, road tax (per vehicle) |
| FINANCING | Full EMI stream, 10% / 60 months |
| ENERGY | Tariff × consumption, net of charging losses |
| MAINTENANCE | Fleet service cost, net of the EV tyre-wear premium |
| INSURANCE | OD premium, actual, EV higher than CNG |
| CHARGING INFRA-STRUCTURE | Charger capex, installation, DISCOM load sanction, N+1 redundancy, management software, AMC, and replacement at year 5 |
| DOWNTIME | Foregone net earnings, decomposed by cause |
| RESIDUAL | Credit at end of horizon, escalated |
| CNG-SIDE INFRA | Periodic cylinder hydro-testing |

## 1. CONFIGURATION USED

| | |
|---|---|
| VEHICLE | MACH1 — BEV, LFP 45 kWh, PMSM 80 bhp continuous, single-speed |
| COMPARATOR | Maruti Dzire Tour S (CNG) |
| DUTY | 200 km/day × 26 days × 312 days = 5,200 km/mo · 62,400 km/yr |
| CONSUMPTION | 12.84 kWh/100 km (verified duty cycle, battery phase |
| CHARGING | Depot AC overnight, 90% wall-to-battery efficiency |
| GROSS FARE | ₹11/km net fare → ₹57,200/month |
| PLATFORM FEE | ₹67/vehicle-day (Ola zero-commission) → ₹1,742/month |
| NET FARE | ₹10.665/km → ₹55,458/month (derived) |
| FINANCING | 10% annual, 60 months, no down payment (deck's assumption) |
| HORIZON | 5 years (loan) and 8 years (battery-phase horizon) |
| DISCOUNT RATE | 12% nominal annual — a fleet operator's hurdle rate |
| ESCALATION | Electricity 4.5% · CNG 3.5% · maintenance 4.0% · insurance 5.0% · residual 4.5% |
| CITIES | Delhi, Mumbai, Bengaluru — different tariffs, road tax and CNG prices |

## 3. THE COST STACK
## DELHI, 5 YEARS, FINANCED (3,4,5)

| COMPONENT | EV | CNG | EV ADVANTAGE |
|---|---|---|---|
| On-road (ex-showroom + GST + road tax) | ₹12,72,600 | ₹8,51,488 | −₹4,21,112 |
| Finance (EMI ₹29,743 / ₹18,092 × 60) | ₹16,22,339 | ₹10,85,496 | −₹6,99,077 |
| Energy | ₹2,00,304 | ₹9,48,480 | +₹7,48,176 |
| Maintenance | ₹93,600 | ₹2,10,000 | +₹1,16,400 |
| Insurance | ₹2,10,000 | ₹1,97,400 | −₹12,600 |
| Residual Credit | −₹3,30,876 | −₹3,40,595 | +₹9,719 |
| Subtotal | ₹17,95,367 | ₹21,00,781 | +₹3,05,414 |
| Charging infrastructure | ₹1,74,000 | ₹4,500 | −₹1,69,500 |
| Total | ₹19,69,367 | ₹21,05,281 | +₹1,35,914 |
| Cost/Km | ₹6.31 | ₹6.75 | +6.5% |

## 4. OPERATOR TCO — THREE CITIES

| CITY | BASIS | EV ₹/km | CNG ₹/km | EV ADV. |
|---|---|---|---|---|
| Delhi | Financed 60 mo | 5.75 | 6.73 | +14.5% |
| Delhi | Cash outright | 4.63 | 5.98 | +22.6% |
| Mumbai | Financed 60 mo | 6.47 | 7.00 | +7.6% |
| Mumbai | Cash outright | 5.35 | 6.32 | +15.3% |
| Bengaluru | Financed 60 mo | 6.66 | 7.25 | +8.2% |
| Bengaluru | Cash outright | 5.45 | 6.51 | +16.4% |

*LEVER 1 — **FINANCING STRUCTURE NEARLY DOUBLES THE ADVANTAGE**. CASH PURCHASE GIVES +17.5% AGAINST FINANCED +8.4%, IN THE SAME CITY, ON THE SAME VEHICLE. THE DRAG IS THE INTEREST ON A ₹14 L ASSET: ROUGHLY **₹3.85 L OF INTEREST**, AGAINST ₹2.33 L ON THE CNG.*

*THAT IS A NEGOTIATION LEVER, NOT A PHYSICS ONE. A FLEET OPERATOR BUYING AT SCALE CAN NEGOTIATE A LONGER TENURE, A LOWER RATE, OR AN EV-SPECIFIC PRODUCT. **A 7-YEAR TENURE MOVES MOST OF THE HORIZON FROM "FINANCED" TO "CASH," AND THAT IS WORTH MORE THAN ANY TARIFF NEGOTIATION WE COULD WIN.***

| HORIZON | EV ₹/km | CNG ₹/km | EV ADV.no infra | With infra | NPV + infra |
|---|---|---|---|---|---|
| 5yr | 5.75 | 6.73 | +14.5% | +6.5% | +6.7% |
| 6yr | 5.06 | 6.34 | +20.1% | - | - |
| 7yr | 4.57 | 6.05 | +24.4% | - | - |
| 8yr | 4.20 | 5.84 | +28.0% | +17.4% | +14.6% |

*LEVER 2 — **HOLD THE VEHICLE LONGER.***
*DELHI, FINANCED. THE CASE ROUGHLY **TRIPLES BY YEAR 8** BECAUSE THE LOAN IS REPAID AND THE ASSET KEEPS EARNING. THIS IS THE STRONGEST SINGLE ARGUMENT IN THE DOCUMENT AND IT IS AN OPERATOR ARGUMENT — IT BELONGS TO WHOEVER OWNS THE VEHICLE.*

## 5.SENSITIVITY-

| INPUT CHANGE | EV ADVANTAGE |
|---|---|
| BASE (DELHI: ELEC ₹4.50, CNG ₹76/KG, 25 KM/KG) | **+14.5%** |
| CNG 22 KM/KG CITY ONLY, AC ON | **+19.5%** |
| CNG 30 KM/KG — HIGHWAY, NOT METRO DUTY | **+7.6%** |
| CNG PRICE ₹76 → ₹90/KG | **+21.1%** |
| CNG RESIDUAL 40% → 45% | **+12.8%** |
| ELECTRICITY ₹4.50 → ₹8.00 (GENERIC COMMERCIAL) | **+7.1%** |
| ELECTRICITY ₹4.50 → ₹10.00 | **+2.9%** |
| ELECTRICITY ₹4.50 → ₹12.00 (PUBLIC DC) | **−1.4% ✗** |
| ELECTRICITY ₹4.50 → ₹18.00 | **−14.1% ✗** |
| EV RESIDUAL 26% → 36% (DECK'S OLD ASSUMPTION) | +20.6% |
| CONSUMPTION 12.84 → 15.38 KWH/100 KM (DECK'S FIGURE) | +12.7% |
| **WORST CASE STACKED (15.38 + ₹12 + RESID 26%)** | **−6.4%** |
| **BEST CASE STACKED (12.84 + ₹4.50 + RESID 36%)** | **+20.6%** |
| 8-YEAR HORIZON | +28.0% |

***THREE CONCLUSIONS STAND OUT:***

1. ***THE EV ADVANTAGE HOLDS UP AGAINST MEASURED CNG EFFICIENCY.*** *THE MEASURED BAND LEAVES AN EV ADVANTAGE OF **+19.5%** TO **+11.6%**, GIVING THE CASE A REASONABLE MARGIN UNDER THESE ASSUMPTIONS.*
2. ***ENERGY PRICES REMAIN THE REAL FRAGILITY.*** *ABOVE ROUGHLY ₹11/KWH, THE **EV LOSES ITS ADVANTAGE.** DEPOT TARIFFS OF ₹4.50–9.50/KWH SUPPORT THE CASE, WHILE PUBLIC DC CHARGING AT ₹12–24/KWH CAN OVERTURN IT.*
3. *UPTIME IS THE BINDING OPERATIONAL RISK. THE **EV HAS 45 HOURS/MONTH OF DOWNTIME** HEADROOM, AGAINST A DOCUMENTED **75 HOURS/MONTH LOST TO CHARGER QUEUES** (GREENSM). THE CASE THEREFORE DEPENDS ON RELIABLE DEPOT CHARGING, NOT JUST VEHICLE ECONOMICS*

| FACTOR | KEY ASSUMPTIONS AND CALCULATIONS | KEY RESULTS / BREAK-EVEN THRESHOLDS | IMPLICATION |
|---|---|---|---|
| 1. Depot charging | Daily energy: 25.68 kWh; 7.4 kW AC charging takes ~3.86 h. Charger cost: ₹50,000 hardware + ₹30,000 installation + ₹10,000 DISCOM connection, increased to ₹99,000/vehicle for N+1 redundancy. Software and AMC: ₹15,000/year. | Depot off-peak tariffs: ₹5.50–9/kWh; public DC retail: ₹18–24/kWh. At ₹4.50/kWh, EV energy costs ₹0.64/km; at ₹8.50, ₹1.21/km; at ₹12, ₹1.71/km. | Depot charging is essential. The report's ₹12/kWh sensitivity turns negative, and public DC retail tariffs are typically higher. Charger infrastructure must be included in TCO. |
| 2. Vehicle uptime | Baseline downtime: 3.5 h/month—0.5 h queues/port unavailability, 1 h planned maintenance, 1.5 h repairs and 0.5 h charger failure. Net cost of lost operating time: ₹200.46/hour. | Baseline loss: ₹702/month, ₹0.135/km or ₹42,097 over five years. Break-even: 25.4 total downtime hours/month, leaving 21.9 additional hours beyond baseline. A reported ~75 h/month without depot charging produces a −28.4% result. | Reliable depot access is a contractual operating requirement, not an automatic property of the vehicle. Charging queues and repairs can erase savings. |
| 3. Battery replacement risk | Main eight-year case assumes no pack replacement. The report's replacement sensitivity uses an earlier nominal, vehicle-only comparison basis. | No replacement: ₹5.75/km, +14.5% EV advantage. ₹4 lakh replacement: ₹7.04/km, −4.5%; ₹8 lakh: ₹8.32/km, −23.5%; ₹12 lakh: ₹9.60/km, −42.6%. | Even a ₹4 lakh replacement reverses the earlier comparison. Battery warranty, capacity-retention guarantees and replacement terms are critical. Recalculate scenarios on the final NPV-inclusive basis before comparing them with the headline result. |
| 4. Battery life and EFC | Battery-side throughput: 8,012 kWh/year. EFC estimates: 178/year using nominal 45 kWh capacity; 181.3 using usable 44.2 kWh; 197.8 using wall energy divided by nominal capacity. | Consistent calculation gives 178 EFC/year and 16.85 years to 3,000 cycles, versus 197.8 EFC/year and 15.16 years on the published basis. Calendar-ageing estimates range from 3.7 years to 80% capacity under a measured linear model to 13.3 years under a √t model at 40°C. | The published EFC figure overstates cycle throughput by 11.1% because it mixes wall energy with battery capacity. Cycle life does not guarantee service life; calendar ageing may dominate. |

MARUTI SUZUKI

# DRIVER EARNINGS

## DRIVER P&L — FLEET-OPERATED VEHICLE

| Monthly line | EV | CNG |
|---|---|---|
| Gross revenue (₹11 × 5,200 km) | ₹57,200 | ₹57,200 |
| less platform access fee (₹67/day × 26) | −₹1,742 | −₹1,742 |
| less energy | −₹3,338 | −₹15,808 |
| less maintenance | −₹1,560 | −₹3,500 |
| less insurance | −₹3,500 | −₹3,290 |
| Take-home | ₹47,060 | ₹32,860 |
| Advantage | +₹14,200 / month | |
| Annualised | ₹5,64,715 | ₹3,94,320 |
| Five-year cumulative | | +₹8,51,976 |

## EV ADVANTAGE IS A FIXED NUMBER

1. *The EV advantage is ₹14,200 every month, at every fare and every duty level. It does not scale with earnings — it scales with* ***nothing.***
2. *So the same ₹14,200 is worth twice as much in percentage to the lowest-paid driver.A driver on a ₹11/km street fare gains 43%. A driver on a ₹22/km airport run gains 16%. Both get the identical rupees.*
3. *And it barely moves with duty — at 150 km/day the EV take-home is 145% of the CNG's, at 300 km/day it is 141%. The* ***ratio is stable.***

| Fare | EV take-home | CNG take-home | Advantage | As % uplift |
|---|---|---|---|---|
| ₹11/km | ₹47,060 | ₹32,860 | ₹14,200 | +43.2% |
| ₹14/km | ₹62,660 | ₹48,460 | ₹14,200 | +29.3% |
| ₹18/km | ₹83,460 | ₹69,260 | ₹14,200 | +20.5% |
| ₹22/km | ₹1,04,260 | ₹90,060 | ₹14,200 | +15.8% |

## WHAT THE PLATFORM TAKES IS IRRELEVANT

| Platform model | EV take-home | CNG takehome | Advantage |
|---|---|---|---|
| Flat ₹67/day (Ola/Uber/Rapido) | ₹47,060 | ₹32,860 | ₹14,200 |
| Legacy 20% per ride | ₹48,799 | ₹34,600 | ₹14,200 |
| Legacy 25% per ride | ₹48,799 | ₹34,599 | ₹14,200 |

## GROSS EARNINGS @ VARYING FARE RATES

| | Arithmetic | Result |
|---|---|---|
| ₹93,600 | ₹18.00/km × 5,200 km/month | ₹93,600 |
| ₹114,400 | ₹22.00/km × 5,200 km/month | ₹114,400 |

## DRIVER P&L — DRIVER-OWNED VEHICLE

| | EV | CNG |
|---|---|---|
| Take-home during the 60-month loan | ₹20,021 | ₹14,768 |
| Take-home year 6 onward | ₹47,060 | ₹32,860 |
| Advantage during loan | +₹5,252 / month | |
| Advantage after loan | +₹14,200 / month | |

## HOW ROBUST IS THE DRIVER CASE?

*The driver advantage is far more robust than the operator advantage. The operator case is sensitive to charger cost and duty; the driver case survives almost everything, because it is a difference between two cost stacks, not a margin that can be squeezed.*

| Stress test | Result |
|---|---|
| CNG price rises to ₹200/kg (today ₹76) | Advantage only then disappears |
| Duty falls from 200 to 150 km/day | +₹10,597 instead of ₹14,200 — still positive |
| Fare halves to ₹5.50/km (legacy commission) | Advantage still ₹14,200 |
| Platform switches to per-ride commission | No change to the advantage |

### WHAT WE WILL NOT CLAIM ON THE DRIVER SIDE

- *Not ₹93,600 in take-home earnings. That is gross revenue; take-home is ₹47,060 at the assumed fare. At ₹11/km, it falls to ₹20,021 if the driver also finances the vehicle.*
- *Not universal driver profitability. Below roughly 60 km/day, one vehicle cannot cover the driver's fixed costs. This is a utilisation problem, not a powertrain problem.*
- *Not better income simply because the vehicle drives better. The economic gain comes from a ₹1.24 lakh/month reduction in operating costs.*

# RIDER REVIEWS

XCELerate 2026

## WHAT RIDERS ACTUALLY COMPLAIN ABOUT

| COMPLAINT | SHARE |
|---|---|
| Driver cancelled after learning destination or payment | 74% |
| Surge pricing | 62% |
| Long waiting times | 53% |
| Unclean vehicles | 30% |
| Driver behaviour | 25% |
| Safety concerns | 6% |

LocalCircles surveyed **17,993 respondents across 325 districts (2025):**
*A separate Bengaluru survey (10,000+ responses) found* ***85%*** *of Uber users and* ***93%*** *of Ola users reported at least one of: cancellations, phone use while driving, surge pricing, unclean cars.*

## WHAT RIDERS SAY ABOUT THE VEHICLE

*Owner reviews repeat four praises — mileage, low maintenance, service-network reach, ride comfort and four complaints: a basic cabin, no infotainment, no rear AC vents, and a 3-cylinder that feels underpowered. One owner put it plainly: "Only perfect for taxi like Ola or Uber, because safety you have to compromise. For personal I would not prefer."*

***There is also a brand problem worth naming: owners of the new-generation Dzire note it is"comfortable and looks premium, but still seen as cabs." The taxi image is the stigma.***

### MARUTI DZIRE TOUR S

| Sub-rating | Score |
|---|---|
| Mileage | 5.0 / 5 |
| Maintenance cost | 4.9 / 5 |
| Performance | 4.8 / 5 |
| Comfort | 4.7 / 5 |
| Safety | 4.1 / 5 |
| Features & styling | 1.2 / 5 |

## WHAT THE RIDER EVIDENCE SAYS WE SHOULD CHANGE

1. *Fix the features gap. The benchmark vehicle scores* ***1.2/5 on features****. An EV programme is exactly when to add what the CNG generation never got — rear AC vents, a real infotainment system, connected features. Riders rate tangibility third, but they rate it, and the incumbent's weakest score is ours to take.*
2. ***Front and rear dashcams*** *as standard. Riders name them as a differentiator on premium EV services. Cheap to specify, directly addresses safety which is the top stated priority.*
3. *The* ***boot is a customer feature****. Airport transfers are the margin segment, and the Dzire Tour S manages 382 L. At 400 L we already lead; we should say so in customer terms, not dimension terms.*

## THE UNCOMFORTABLE CUSTOMER FINDING

*The rider's top complaint is* ***cancellation****, and our vehicle* ***cannot fix*** *it on its own.. A driver cancels because the trip is unprofitable or the destination is awkward. Depot charging and a ₹99,000 charger that stays working make the driver more reliable , that is a real causal chain , but the honest position is that vehicle design moves rider satisfaction at the margin, and the large* ***win is in fleet operations.***

## WHERE MACHI WINS AND LOSES

| Rider priority | Evidence | Our position |
|---|---|---|
| Reliability | Top SERVQUAL predictor; 74% top complaint is cancellation | Strongest claim. A 21.9 h/month downtime slack and a depot-charging operating model are the mechanism — a vehicle that is not queueing for a charger is a vehicle that shows up |
| Tangibility / condition | β=.182 predictor; 30% complain of unclean cars | Neutral-to-good. Silence and consistent cabin spec are a genuine upgrade; features are where the Dzire scores 1.2/5, and an EV is the moment to fix that |
| Empathy / courtesy | β=.263 predictor | Operator-controlled, not vehicle-controlled. Driver economics feeding §4 is the indirect lever |
| Price fairness | Explains 47.6% of satisfaction variation | EV caps operator cost but does not guarantee a fairer fare to the rider — this needs a platform commitment, and we should not claim it as a vehicle feature |
| Safety | Top stated priority; 6% formal complaints | Already specified: L2 ADAS, airbags, ABS. Note the fleet reality — rear seatbelt compliance is a behaviour problem, not a hardware one |
| Quiet / comfort | Most-praised EV attribute | Free win. An EV sedan is quieter than the CNG sedan it replaces, in the segment where noise is already the top complaint |

MARUTI SUZUKI

# AI & CONNECTED MOBILITY

XCELerate 2026




## TIRE PRESSURE MONITORING SYSTEM

- Continuous Monitoring — Pressure and temperature measured per wheel in real time, displayed live on dashboard with per-wheel status.
- Three-Level Threshold — Normal: 32–35 psi, 30–50°C. Warning: below 28 psi or above 60°C. Critical: below 22 psi or above 70°C — VCU alert triggered at each level.
- Leak Detection — Rate of pressure drop monitored continuously. Rapid or gradual loss flagged before reaching critical threshold, logged to VCU and cloud.
- System Value — Correct pressure reduces rolling resistance and extends range. Early detection prevents mid-shift tyre failures on revenue-earning vehicles.

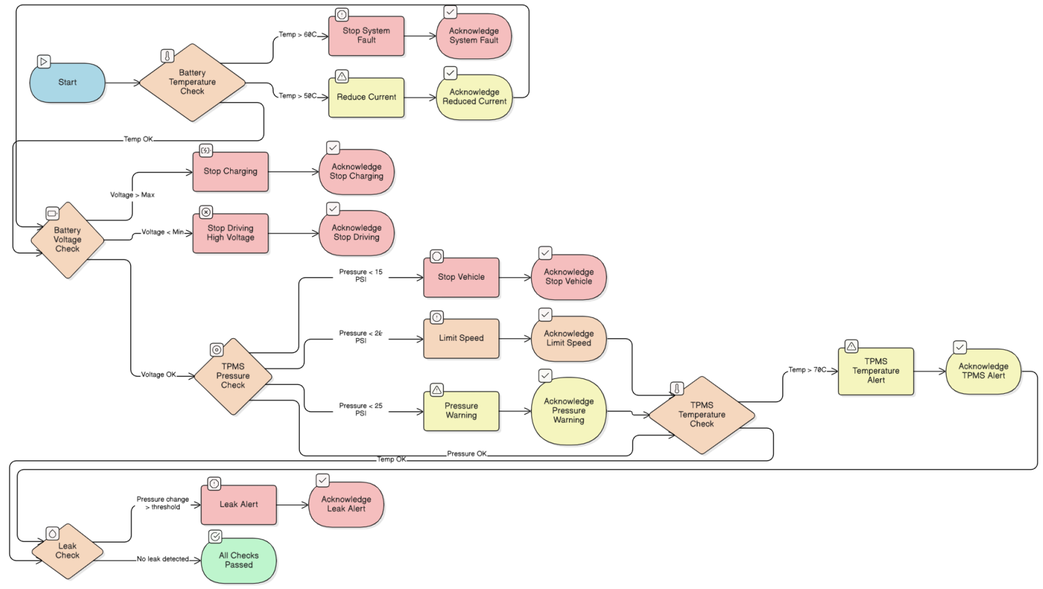



## FAST CHARGING SYSTEM

- Initialization & Safety Check — System verifies battery connection, reads voltage, current and temperature. Charging stops automatically if any parameter is outside safe limits.
- CC Phase — Fixed high current delivered until battery reaches peak voltage. Charges to ~70–80% capacity quickly.
- CV Phase — Voltage held constant, current tapers down to threshold. Prevents overcharging and thermal stress on cells.
- Continuous Monitoring — Parameters tracked throughout. All safety checks remain active. Data logged to VCU and cloud in real time.

## INTEGRATED BATTERY & TYRE SAFETY MANAGEMENT

- Battery Safety: Temperature and voltage thresholds trigger current reduction, charging shutdown, or vehicle cutoff.
- Tyre & Leak Monitoring: TPMS checks pressure, temperature, and sudden pressure drops; unsafe conditions trigger alerts, speed limits, or vehicle stop.
- Fault & Reset Logic: All faults require acknowledgement before reset; checks continuously repeat until all conditions pass.

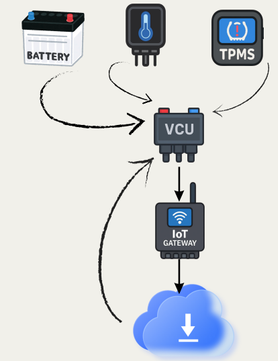



# SENSE. THINK. ACT.

*System Workflow*

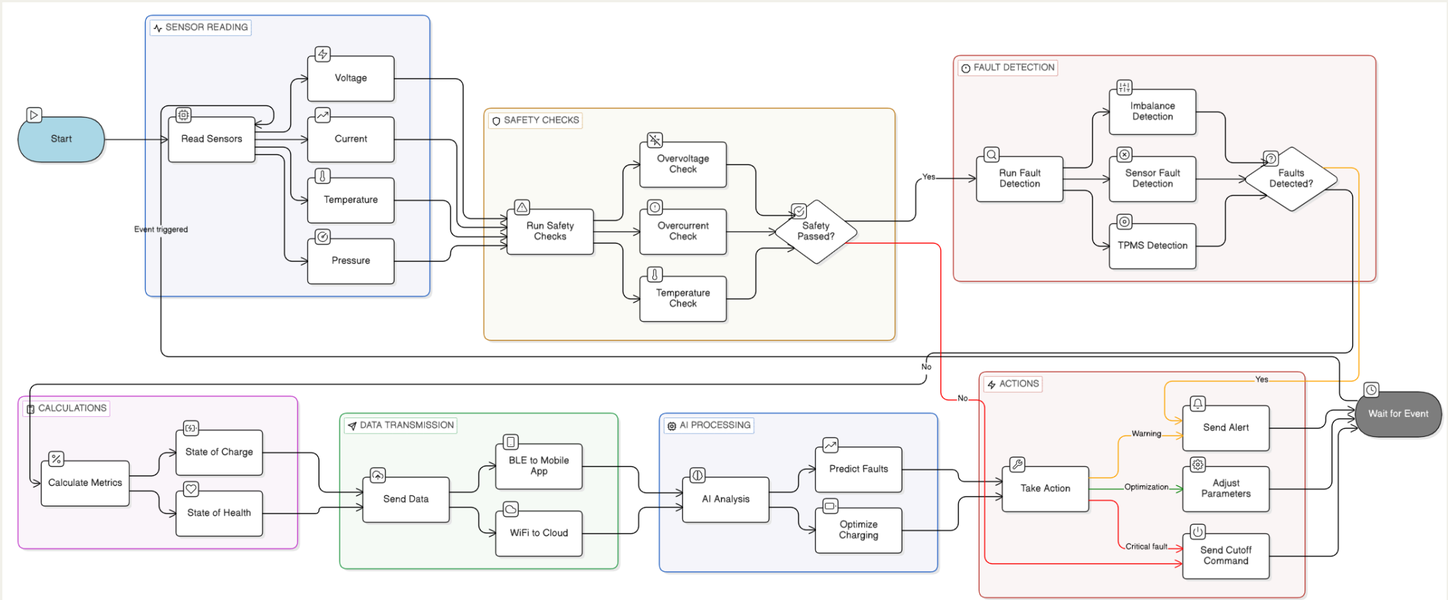
1. *Sensor Acquisition & Safety Checks*
   - *Read voltage, current, temperature and pressure on each trigger.*
   - *Check overvoltage, overcurrent and temperature limits. If any check fails, bypass AI and immediately initiate the appropriate safety action.*
2. *Fault Detection, Calculations & Data Transmission*
   - *In parallel, detect cell imbalance, implausible sensor readings and abnormal tyre-pressure drops; calculate SOC using current integration and*
   - *SOH using charge-discharge history.*
   - *Transmit data via BLE to the driver's app and Wi-Fi to the cloud for fleet monitoring.*
3. *AI-Based Analysis*
   - *Fault Prediction: Use historical patterns and current readings to identify developing faults, such as cell degradation, temperature drift and bearing wear.*
   - *Charging Optimisation: Adapt charging parameters using SOC and SOH, including lowering the constant-current (CC) limit for degraded cells.*
4. *Action & Repeat*
   - *Respond with a warning, real-time parameter adjustment (charging current or motor torque), or hard cutoff for critical faults.*
   - *Return to the wait state and repeat the cycle on the next trigger.*

## SYSTEM ARCHITECTURE




## SMART FEATURES

- Fleet Monitoring — live dashboard per vehicle: SOC, location, faults, battery status
- Predictive Maintenance — early fault detection, **≥7-day advance warning, 20–30% less downtime**
- Smart Dispatch — SOC and location matched to demand, **~15% fewer deadhead kilometres**
- Smart Charging — schedule built around shift demand and tariff window, **15–18% lower charging cost**
- Remote Diagnostics — faults identified without physical inspection, reduces workshop time



MARUTI SUZUKI

# ROADMAP

XCELerate 2026

MACH1 · FLEETLINE E1

## Roadmap — gated, not dated

FIVE GATES · THREE PHASES

**Measurements decide each step**

1 **Phase 1** — 2026–27 · Delhi NCR

G **Gates G1–G3** — cell · duty · battery

2 **Phase 2** — 2028–30 · Delhi + Bengaluru + Mumbai

G **Gate G4** — scale

3 **Phase 3** — 2031–35 · Pan-India

G **Gate G5** — cells

### PHASE 1

2026–27 · Delhi NCR

- **1,000–2,000 vehicles**
- Margin segment: airport transfers + corporate commute
- 1 × 7.4 kW AC port per vehicle, 10% N+1 spares, ₹99,000/vehicle installed
- Overnight window 25.68 kWh ÷ (7.4 kW × 0.90) = 3.86 h — one port is enough, no DC needed
- Cells: Reliance Jamnagar LFP primary, qualified import as bridge, dual-sourced from day one
- Power: ₹4.50/kWh Delhi BEE EV tariff

*1,000–2,000 is indicative, gate-conditional — not a forecast.*

### PHASE 2

2028–30 · Delhi + Bengaluru + Mumbai

- All three cities clear at five years on the published basis (see ECONOMICS)
- **Cumulative 15,000–25,000 vehicles**

### PHASE 3

2031–35 · Pan-India

- **100,000+ cumulative**

### G1 · CELL

Two qualified LFP sources at price parity within 10%.

**FAIL → Stay dual-imported; do not commit Phase 2 volume.**

### G2 · DUTY

12 months of measured paid km/day ≥ 200.

**FAIL → Re-run the TCO before scaling; the whole case is duty-sensitive.**

### G3 · BATTERY

12 months of per-cell SoH telemetry closing the 3.6× disagreement in published LFP time-to-80% (Naumann √t gives 13.3 yr at 40 °C; Lewerenz's measured linear law gives 3.7 yr).

**FAIL → Capacity-retention guarantee is unsignable.**

### G4 · SCALE

20,000 cumulative vehicles with the 21.9 h/month downtime slack maintained against a 25–40% public-charger session-failure rate.

**FAIL → Downtime slack is consumed; hold the fleet at Phase 2 volume.**

### G5 · CELLS

Measured operating domestic LFP capacity >20 GWh with automotive qualification. A measurement, not a date.

**FAIL → Domestic content stays at L2; the import bridge continues.**

### LOCALISATION

**L1 pack assembly 2026–27** → **L2 domestic cells for a share 2028–30** → **L3 CAM/AM/separator content 2031+**

L3 only if gated. The honest ceiling: ACC PLI has delivered 1.4 GWh of 50 GWh — 2.8%. Reliance Jamnagar is the LFP player (40 GWh, ramping H2 2026); Ola Cell is the only Indian producer at scale and is NMC-first.

### INFRASTRUCTURE

One AC port per vehicle. AIS-138 Part 2 amended 2022 to mandate CCS2 for four-wheelers; OCPP 1.6J.
Beyond ~300 km one-way, destination charging is mandatory — a fleet-contract requirement, not a vehicle spec.
Range envelope: 268 km at 15% reserve (315 km at zero).
Feasible: Delhi–Agra 200 · Mumbai–Pune 150 · Mumbai–Nashik 170 · Bengaluru–Mysuru 145. Marginal: Delhi–Jaipur 280 (88.8% of usable).

### POLICY

| STATE | WHAT IT GIVES | EXPIRY |
|---|---|---|
| **Delhi** | 0% road tax for life of vehicle; ₹1 lakh scrapping incentive. No purchase incentive for electric cars. | **31-Mar-2030** |
| **Maharashtra** | 15% of ex-factory, capped ₹2 lakh, for transport e-4W — 25,000 units; 100% toll exemption on Mumbai–Pune Expressway and Samruddhi Mahamarg; EV road-tax exempt. | **31-Mar-2030** |
| **Karnataka** | 20–25% capex, stamp-duty reimbursement, plug-and-play clusters. No consumer incentive. | **2030** |
| **Central** | PM E-DRIVE has no e-4W category at all. | **—** |

**Both Delhi and Maharashtra expire five years before the 2035 target.**

### ECONOMICS

NPV at a 12% hurdle, escalated, financed, including the ₹99,000 charger.

| | DELHI | MUMBAI | BENGALURU |
|---|---|---|---|
| **5-yr** | **+6.7%** | **+4.0%** | **+0.9%** |
| **8-yr** | **+14.6%** | **+10.6%** | **+9.0%** |
| **2035, 5-yr** | **+15.7%** | **+10.4%** | **+9.0%** |
| **2035, 8-yr** | **+23.1%** | **+16.4%** | **+16.3%** |

Power-escalation break-even at five years: Delhi 14.5%/yr, Mumbai 8.8%, Bengaluru 8.7%. Electricity escalation is unambiguously bad for the EV in every city (worst Mumbai −9.2 pp); the CNG leg is what carries it.

**Only one cell in the entire 48-cell model is negative — Bengaluru, 5-yr, cash purchase, −0.4%.**

MARUTI SUZUKI

# SYNTHESIS

XCELerate 2026

MACH1 · FLEETLINE E1

## The argument, in six steps

SYNTHESIS

**DELHI · 200 KM/DAY**

1

### THE PROBLEM

Indian ride-hailing fleets run on CNG — not by preference, because nothing was built for their duty cycle. Fleet duty is 200 km/day, 312 days a year, return-to-base every night. It is the most demanding duty cycle an EV can have, and the one the Indian market has served worst.

2

### THE INSIGHT — THE PIVOT

Volume and margin live in different segments, and they need different vehicles.

**VOLUME** NOT THE LAUNCH SEGMENT

- street-hail intra-city · ~58% of rides
- unpredictable dispatch · price sensitivity 0.85 · ~10% EV adoption

**MARGIN** THE LAUNCH SEGMENT

- **airport + corporate commute · ~19% combined**
- **144–210 paid km/day · pre-booked | · price sensitivity 0.30–0.35 · ~18% EV adoption**

**We did not chase the largest segment. We launched where the economics close and EV adoption already exists.**

3

### THE VEHICLE

BEV · LFP 45 kWh · PMSM 80 bhp · **268 km at 15% reserve (315 km at zero) · 3,995 mm** — deliberately 5 mm inside the small-car tax line.
Designed around one number: **200 km/day**, return-to-base, one **7.4 kW AC port**, overnight.

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### THE MATH

NPV at a **12%** hurdle, escalated, with the **₹99,000/vehicle depot charger** inside the cost boundary.

| | DELHI | MUMBAI | BENGALURU |
|---|---|---|---|
| 5-year | +6.7% | +4.0% | +0.9% |
| 8-year | +14.6% | +10.6% | +9.0% |

All three clear at five years. One cell in the whole 48-cell model is negative — Bengaluru, bought cash, **−0.4%.**

Broker and platform fees are identical on either powertrain, so the EV-vs-CNG difference is unaffected by how commission is structured.

**Inter-city:** Delhi–Agra 200, Mumbai–Pune 150, Mumbai–Nashik 170, Bengaluru–Mysuru 145 all fit inside the 268 km envelope. Delhi–Jaipur 280 is marginal. Beyond ~300 km, destination charging is a fleet-contract requirement, not a vehicle spec.

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### WHY LFP, AND WHAT WE ADMIT

Published LFP studies disagree **3.6×** on time-to-80%. So per-cell State-of-Health telemetry is specified, **not optional** — it is the mechanism that makes an 8-year capacity-retention guarantee credible.

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### THE RISK WE CANNOT HEDGE

No central scheme pays for an electric passenger car. PM E-DRIVE's ₹11,900 cr covers 2W, 3W, rickshaws, ambulances, trucks, buses and charging — not 4W cars.
Maharashtra does pay: **15% of ex-factory, capped ₹2 lakh**, for transport e-4W, 25,000 units.

> **And the concessions we do rely on — Delhi's life-of-vehicle road-tax exemption, Maharashtra's e-4W incentive — both expire 31 March 2030, five years before the target year. Renewing them is a policy position worth about five points, not an engineering one.**

**1,000–2,000 vehicles, Delhi,** gated on cell supply · measured duty · twelve months of our own battery data.